Kali Linux Cybersecurity Lab Setup

NetworkWalks Internship | Practical Task Documentation

1. Objective

To set up and configure a Kali Linux virtual machine using Oracle VirtualBox and establish a basic cybersecurity lab environment for hands-on learning.

2. Environment

- Host Operating System: Windows 11
- Virtualization Platform: Oracle VirtualBox 7.2.4
- Guest Operating System: Kali Linux 2026.3
- Allocated RAM: 4 GB
- Network Mode: NAT Network

3. Tasks Completed

3.1 Kali Linux Virtual Machine Setup

- Created and configured a Kali Linux virtual machine.
- Allocated system resources.
- Successfully booted the virtual machine.

3.2 Network Configuration

- Created a VirtualBox NAT Network.
- Configured static IP addressing inside Kali Linux.
- Configured DNS settings.
- Verified network connectivity.

3.3 Connectivity Verification

Connectivity checks were performed for gateway reachability, external IP connectivity, and DNS resolution. All performed connectivity tests were successful.

3.4 Nmap Installation

- Installed Nmap in Kali Linux.
- Verified the installation using the version command.
- Prepared the environment for authorized network scanning exercises.

3.5 Shared Folder Configuration

- Configured a shared folder between Windows and Kali Linux.
- Enabled file access between the host and guest systems.
- Verified shared folder visibility from Kali Linux.

3.6 Virtual Machine Snapshot

- Created a baseline snapshot of the Kali Linux environment.
- Used the snapshot as a recovery point for future lab activities.

4. Tools and Technologies

- Kali Linux
- Oracle VirtualBox
- Nmap
- Linux Terminal
- NetworkManager
- NAT Networking

5. Learning Outcomes

- Linux virtual machine management.
- Basic network configuration.
- Static IP and DNS configuration.
- Network connectivity troubleshooting.
- Installing and verifying cybersecurity tools.
- Host-to-guest file sharing.
- Virtual machine backup and recovery using snapshots.

6. Privacy and Security

- Actual IP addresses and gateway details are intentionally excluded.
- Personal usernames and host-specific file paths are not disclosed.
- No credentials, passwords, or sensitive information are included.
- Network testing is limited to authorized lab environments.

7. Conclusion

Successfully established a Kali Linux cybersecurity lab environment using Oracle VirtualBox as part of the NetworkWalks internship task. This setup provides a foundation for further practical learning in network security, reconnaissance, and authorized Nmap scanning.

8. Next Steps

- Perform authorized Nmap scanning in a controlled lab.
- Explore network reconnaissance techniques.
- Practice basic network security analysis.
- Document practical findings and observations.

Internship: NetworkWalks **Task:** Kali Linux Cybersecurity Lab Setup

Status: Initial lab configuration completed